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APRIL 12, 1960
SUPERSEDING
Int. Fed. Spec. TT-E-00776a (GSA-FSS)
July 29, 1959 and
Fed. Spec. TT-E-776
March 9, 1949

FEDERAL SPECIFICATION

ETHYLENE GLYCOL MONOBUTYL ETHER (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers ethylene glycol monobutyl ether (for use in organic coatings) (See 6.1).

1.2 Classification.

1.2.1 *Type and grade.*—This specification covers one type and one grade of ethylene glycol monobutyl ether.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-636—Box, Fiberboard.
PPP-C-96—Cans, Metal, 28 Gage and Lighter.
PPP-D-729 — Drums, Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-D-760—Drums and Pails, Metal (5 and 16.64 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 — Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 **Other publications.**—The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

American Society for Testing Materials Standards:

D–1209—Test for Clear Liquids (Platinum Cobalt Scale.

D–1353—Nonvolatile Matter in Lacquer Solvents and Diluents.

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 **Quantitative requirements.**—The ethylene glycol monobutyl ether shall conform to the requirements of table I.

TABLE I.—*Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Specific gravity at 20°/20°C.	0.900	0.905
Nonvolatile matter from 100 ml. (gram)	---	0.005
Acidity (mg. of KOH per gram of sample)	---	0.1
Distillation:		
Initial boiling point at 760 mm. pressure (°C.)	165	---
Distillate below 167°C. (percent by volume)	---	5
Dry point at 760 mm. pressure (°C.)	---	173
Distillate below 171°C. (percent by volume)	95	---
Color-platinum cobalt scale	---	15

3.2 **Qualitative requirements.**

3.2.1 *Appearance.*—The ethylene glycol monobutyl ether shall be clear and free from sediment and suspended matter when examined by transmitted light.

3.2.2 *Odor.*—The material shall have the characteristic odor of ethylene glycol monobutyl ether and leave no residual odor after drying from filter paper.

3.2.3 *Water.*—The ethylene glycol monobutyl ether shall be miscible without turbidity with 19 volumes of 10° heptane at 20°C.

3.2.4 *Sulfur compounds.*—The ethylene glycol monobutyl ether shall be free from hydrogen sulfide and sulfur dioxide.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Sampling and inspection.**—Unless otherwise specified, sampling and inspection shall be in accordance with the applicable methods 1011, 1021 and 1031 of Federal Test Method Standard No. 141.

4.3 **Test Procedures.**—Tests shall be made in accordance with the applicable methods of Federal Test Method Standard No. 141.

4.3.1 The following tests shall be conducted in accordance with Federal Test Method Standard No. 141.

Test	Method
Appearance	4261
Water	4083
Specific gravity	4183
Acidity	5252
Distillation	4303
Sulfur	5311
Odor	4401

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4.4 **Color.**—Determine the color in accordance with ASTM Method D–1209.

4.5 **Nonvolatile matter.** — Determine the nonvolatile matter in accordance with ASTM Method D–1353.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Unless otherwise specified, ethylene glycol monobutyl ether shall be packaged in 1-quart or 1-gallon cans conforming to type V, class 4, oblong, of Federal Specification PPP–C–96; in 5-gallon drums conforming to type I, class 1, of Federal Specification PPP–D–760; in 55-gallon drums conforming to type II of Federal Specification PPP–D–729; or shall be shipped bulk.

5.1.2 *Level C.*—Unless otherwise specified, ethylene glycol monobutyl ether shall be packaged in containers conforming to the Consolidated Freight Classification Rules or other regulations applicable to such supplies.

5.2 Packing.

5.2.1 *Level A.*—Unless otherwise specified, twelve 1-quart, or six 1-gallon cans shall be packed in a fiber box conforming to type I or II, class 2, W5C or W5S of Federal Specification PPP–B–636. The 5- and 55-gallon drums require no further packing.

5.2.2 *Level B.*—Unless otherwise specified, packing shall be in accordance with 5.2.1 except that fiber boxes shall conform to type I or II, class 1 of Federal Specification PPP–B–636.

5.2.3 *Level C.*—Unless otherwise specified, ethylene glycol monobutyl ether shall be packed in containers conforming to the Consolidated Freight Classification Rules or other regulations, for safe delivery to destination at the lowest applicable rate.

5.3 Marking.

5.3.1 *Civil agencies.* — Marking for civil agencies shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—Marking for military agencies shall be in accordance with Military Standard MIL–STD–129.

5.3.3 *Precautionary marking.*—All individual containers shall be marked with the following precautionary marking:

CAUTION! VAPOR HARMFUL.
KEEP AWAY FROM HEAT AND FLAME.
USE ONLY WITH ADEQUATE VENTILATION.
AVOID PROLONGED BREATHING OF VAPOR.
AVOID PROLONGED OR REPEATED SKIN CONTACT.

6. NOTES

6.1 **Scope and intended use of material.** — Ethylene glycol monobutyl ether covered by this specification is intended for use as a solvent in organic protective coatings, particularly cellulosic lacquers and lacquer thinners, as well as quick drying varnishes and enamels. It has a low rate of evaporation and a mild odor.

6.2 Ethylene glycol monobutyl ether should be purchased by volume, the unit being a U. S. gallon of 231 cubic inches at 20°C. (68°F). The volume of deliveries may be determined by dividing the net weight, in pounds, by the weight per gallon. To obtain the weight per gallon, multiply the specific gravity at 20°/20°C. by 8.322. One gallon of ethylene glycol monobutyl ether at 20°C. (68°F). weighs between 7.490 and 7.531 pounds.

6.3 As a matter of information, the specific gravity range of material conforming to this

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specification (0.900 to 0.905 at 20°/20°C) is approximately equivalent to a range of 0.903 to 0.908 at 15.56°/15.56°C. and a range of 0.897 to 0.902 at 25°/25°C.

6.4 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Level of packaging and packing required (see 5.1 and 5.2).

(*c*) Size of package and quantity required.

(*d*) Details of inspection and test records required (see 4.1).

6.5 Transportation description.—Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Ether, ethyl, other than anesthesia ether. (Released valuation not to exceed 50 cents per pound).

Carload minimum weight 30,000 pounds.

Motor:

Ether, ethyl, other than anesthesia ether. (Released valuation not to exceed 50 cents per pound).

Truckload minimum weight (W)30.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O
Navy—W
Air Force.

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